Distinguishing graded plantarflexor hyperreflexia from graded dorsiflexor weakness in simulated post-stroke gait: a search over 81 sagittal kinematic variables

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Running title. Distinguishing calf hyperreflexia from dorsiflexor weakness in simulated gait

Abstract

Background. Apportioning a spastic equinovarus foot between calf overactivity and dorsiflexor weakness decides between opposite treatments, and the procedure that does it, a diagnostic nerve block, is invasive because no non-invasive measurement apportions. Which gait reading carries that apportionment has not been asked where both magnitudes are known.

Methods. We lesioned a 19-degree-of-freedom, 22-muscle reflex-controlled model by a velocity-dependent stretch reflex on soleus and gastrocnemius and by scaling of tibialis anterior force, re-optimising the controller for each lesion so both magnitudes are known exactly. Eleven lesion conditions and one unlesioned control condition passed a fixed admissibility criterion; a separate 84-cell grid crossed the lesions. We scored 81 candidate sagittal variables rather than assume an endpoint.

Results. Peak ankle dorsiflexion in swing separated the lesions completely, run-level gap 6.265° or 175 times the largest measured artefact, and graded the reflex at $\rho = -1.000$ across five gains. Seventeen runs with a third lesion dying in the same band read as unimpaired, so short survival does not produce the signature. Under 3° of jitter they stayed distinguished at 0.995 accuracy against the knee's 0.939. The asymmetry runs through the scan, though its size depends on scoring: across three measurement-error thresholds and three rank conventions, 14 to 28 of the 81 candidates grade calf overactivity usably against 1 to 7 for weakness, a ratio that never inverts and that lies between 3.7 and 14 across those readings, although it is unbounded above under a stricter common threshold; under this study's noise model the weakness count falls to zero for the three variables tested. On the mixed grid none of four combinations tested recovers weakness without first knowing the tone, the weakness signal reversing across the gain range.

Conclusions. Moving from the knee at contact to the ankle in swing raises the standardised effect against the spread a stroke cohort presents from $d = 0.59$ to 1.50 and cuts misassignment of a single reading from 38 to 23 per cent, on a variable gait laboratories already record. The figures calibrate a group and do not constitute a test for an individual patient. Both magnitudes are known here and in no patient, the reading grades calf overactivity and not its mechanism, the weakness series stops at ×0.80 where patients reach lower, and one patient cohort reports a dorsiflexor-force association of the size this model calls faint.

Keywords. stroke; spasticity; muscle weakness; gait analysis; musculoskeletal modelling; equinus

1. Background

Spastic equinovarus foot after stroke has three treatable causes: calf spasticity, shortening of the triceps surae–Achilles complex, and dorsiflexor weakness. They take opposite treatments, chemodenervation or neurotomy for overactivity, stretching or lengthening for shortening, and orthosis, stimulation or tendon transfer for weakness, and the consequences of assigning them wrongly are asymmetric [1]. Botulinum toxin injected into a triceps surae that is weak rather than overactive reduces the plantarflexor moment available for knee stability in stance, and ref [1] describes botulinum toxin as producing a reversible blockade lasting three to six months; selective neurotomy and tendon lengthening are not reversible, and the block is used most often before those procedures. Published practical guidance calls the diagnostic motor nerve block "a mandatory step" in deciding which cause a given limb has, and uses it alongside clinical examination rather than in place of it [1]. However, the block is itself invasive, and it is used because no non-invasive measurement apportions.

Bedside tests address parts of the problem. Slow-versus-fast stretch (Tardieu R1 versus R2) estimates the dynamic component, and the Silfversköld manoeuvre separates gastrocnemius from soleus restriction by comparing passive dorsiflexion with the knee extended and flexed. Both are approximations, and the guidance keeps the block alongside them and does not replace them with it, without stating its reason [1]. A retrospective series of 50 chronic stroke patients supports the block as a screening tool before botulinum toxin, although the block itself improved every outcome more than the subsequent injection did and therefore overestimates what the toxin will achieve [2]. In 40 adults with brain or spinal-cord injury, instrumented gait analysis and the tibial block carried distinct information and proved complementary rather than interchangeable [3].

We do not address any of the block's purposes directly. Varus is not modelled, and the block population is predominantly equinovarus, in which tibialis anterior is often itself overactive rather than simply weak [3, 20]; the scope here is sagittal-plane equinus only. The components also co-occur. In 42 stroke patients with dynamic ankle spasticity, affected dorsiflexor strength had a median manual muscle test grade of 1 (IQR 0 to 4) against 5 (3 to 5) in propensity-matched stroke patients without spasticity ($p < 0.001$), and affected *plantarflexor* strength was reduced in the same limbs (4 versus 5, $p < 0.001$) [4]. The clinical question is therefore usually how much of each cause is present.

Two groups have measured the ankle at contact in this population. Ouyang et al. measured ankle angle at four gait events, initial contact, toe-off, maximum dorsiflexion and maximum plantarflexion, and no knee kinematics at all; ankle angle at contact differed between the spastic and matched non-spastic groups, and of five variables entered into their model and four retained, the two that remained *significantly* associated with spasticity were bedside measures and not gait variables, namely passive plantarflexion range and dorsiflexor strength [4]. Mendes-Andrade et al. grouped 34 chronic hemiparetic patients by forefoot versus rearfoot contact and found ankle-at-contact differing between groups (−3.67° versus +2.00°, $p = 0.003$), a grouping variable that is itself ankle position at contact [5]. Neither study, therefore, reports a gait variable that carries the apportionment.

The knee has been read at the same instant with the opposite outcome. Mendes-Andrade et al. report knee angle at contact (8.30° versus 8.03°, $p = 0.929$) descriptively [5], although in 30 ambulatory children with spastic cerebral palsy both knee flexion at contact and maximal knee extension in stance correlated positively with graded hamstring spasticity on the Modified Tardieu Scale [6], which establishes that knee-at-contact can index the velocity-dependent component of spasticity in a muscle crossing the knee directly. Only one study has attempted the equivalent reading in adult stroke: a pilot relating graded ankle tone to knee angle through stance in 12 subacute hemiparetic adults using 2D video [7]. Its positive finding is a dorsiflexor one, tibialis anterior tone correlating with knee angle at initial contact ($\rho = 0.662$, $p = 0.019$), while gastrocnemius tone, the muscle class lesioned here, was not significantly related to knee angle at any stance phase ($p = 0.22$, then 0.45, 0.24, 0.48 and 0.85). The machine-readable rendering of that paper's coefficient table which we retrieved is internally inconsistent at n = 12, and we could not obtain a clean copy, so we rely on the p values alone; only nine of the 12 participants had any gastrocnemius tone.

Where both components have been measured together, they move the same joint. In 30 chronic stroke patients, peak stance knee hyperextension related weakly to gastrocnemius spasticity and moderately to dorsiflexor and knee-flexor strength [8], and plantarflexor weakness is separately associated with the same deviation (n = 20, p = 0.044) [17]. Both are associations at an event rather than discriminations between causes. A six-patient series found dynamic EMG changing the treatment hypothesis in post-stroke equinus, two patients whose clinical assessment agreed showing different swing-phase patterns and focal inhibition being withdrawn in one, and it documents the barriers keeping that method out of routine use [20]. Where contractures of different muscles are distinguished, in ten healthy adults wearing an exoskeleton, the knee is a *common* feature rather than the discriminating variable, discrimination being carried by the hip, pelvis and ankle [18].

Simulation can set a deficit of known magnitude and read what follows. Reflex-gain forward simulation reproduces hemiparetic-like kinematics under gastrocnemius hyperreflexia, so the event examined here is not new, and it reaches this study's endpoint as well: reduced suppression of spindle feedback during swing contributes to increased plantarflexion and so to hampered toe clearance, a deficit the literature attributes mainly to pretibial weakness with the role of hyperreflexia not specifically considered [9]. The qualitative claim that calf hyperreflexia can reduce swing dorsiflexion is therefore not ours. What that paper does not supply is a magnitude: its gain factors were chosen arbitrarily, so by its own statement no claim about absolute effect can be made, it simulates no weakness, and it proposes no discriminator. Plantarflexor weakness and contracture have been simulated with per-condition re-optimisation [10], plantarflexor hyperreflexia has been contrasted against plantarflexor weakness [11], and ground-truth impairments propagated to kinematics in personalised cerebral-palsy models explained on average 17 per cent of deviation and never more than 39, contracture having the largest single effect [12]. However, each of these introduces one impairment at a time, so none of them addresses an apportionment.

We lesioned a reflex-controlled musculoskeletal model in two ways, by a velocity-dependent stretch reflex on soleus and gastrocnemius and by scaling of tibialis anterior force, with the controller re-optimised for each lesion so that both magnitudes are known exactly. We then scored 81 candidate sagittal variables across three joints and the whole gait cycle, in degrees and against a published motion-capture error, to establish which of them separates the two lesions, which grades each of them, and what remains recoverable when both are present. The two severity series are themselves unmatched in a way that bears on the comparison, which §4.4 returns to.

2. Methods

2.1 Model, controller, optimisation

3D model `H1922v7b3`: 19 degrees of freedom, 22 muscles, pelvis to femur to tibia to calcaneus, in SCONE 2.4.4.3333 with Hyfydy 1.12.6.1412. A reflex controller of the Geyer and Herr type [16], their positive and negative force and length feedback laws extended from the original planar seven-muscles-per-leg formulation to this model's 22 muscles, with gains optimised by CMA-ES over 90 generations from a converged unlesioned parameter file, 20 s maximum duration and a 5 ms control step, at six seeds per condition except at the two deepest reflex gains, which we ran with four.

The objective is a composite of five terms: a gait term (weight 100) with a minimum velocity of 1.0 m/s and a termination height of 0.85; cubed muscle activation (1000); metabolic effort on the Wang 2012 model as cost of transport (0.1); ground-reaction force above 1.3 body weights (10); and joint-limit penalties on ankle and knee (0.1 each). No term rewards toe clearance, so nothing in the objective pushes the weakness group to compensate for a dropped foot by exaggerated hip and knee flexion; that absence follows from the objective and is not a result. We re-optimised the controller for every lesion, so each analysed gait is that lesioned system's own best solution, and reached severe conditions by warm-start chaining, seeding each severity level from the previous level's optimum, following Ong et al. [10].

2.2 Lesions

Hyperreflexia: a `MuscleReflex` on left soleus and gastrocnemius, velocity gain KV, delay 0.020 s, `allow_neg_V = 0`, with maximum isometric forces unchanged (soleus 3549 N, gastrocnemius 2241 N). The velocity feedback is an addition to the Geyer–Herr formulation, not one of its laws. **Weakness**: scaling of `tib_ant_l` maximum isometric force from its base 1759 N, at ×0.80, ×0.87, ×0.892, ×0.90, ×0.915 and ×0.95. We imposed the two lesions separately, so the hyperreflexia group carries a calf at full strength, which no hemiparetic calf is [1, 4]; §4.4 returns to this. Gains below are delivered, measured directly in the model file as a multiplier of $2.000000 \pm 7 \times 10^{-6}$ over 102 measurements, and the archived outputs are labelled with nominal values, exactly half:

delivered KV	**0.025**	**0.050**	**0.100**	**0.140**	**0.200**	**0.220**	**0.240**
archived label	0.0125	0.025	0.050	0.070	0.100	0.110	0.120

2.3 Admissibility

A run is one lesioned simulation under one optimisation seed. Criterion G requires a simulation end time of at least 9.73 s and at least five complete gait cycles inside [1.00, 9.73] s, with the last kept cycle dropped. Foot contact is the rising edge of `leg0_l.grf_norm_y` through 0.05, and every angle reported at contact is the value at that sample averaged over the kept cycles, a mean of 5.3 cycles per run.

Twenty-three hyperreflexia runs, 36 weakness runs and 6 control runs pass, from five hyperreflexia conditions, six weakness conditions and one control chain. A separate 84-cell mixed grid, 74 of them admitted, crosses four tibialis anterior scalings (×0.60, ×0.70, ×0.80 and ×1.00) with four delivered reflex gains (0.025, 0.050, 0.100 and 0.220) and supports §§3.4 and 3.7. Attrition falls entirely on the hyperreflexia group and is not monotone in gain: 6 of 6 at KV 0.100 and 0.140, 4 of 6 at 0.200, 4 of 4 at 0.220 and 3 of 4 at 0.240. The earliest hyperreflexia failure is at 9.81 s, only 0.08 s above the threshold, which makes the threshold's provenance matter; it was fixed in registrations predating these runs by several hundred optimisation rounds. Two protocol artefact values are used, 0.586° for knee angle at contact and 0.036° for peak dorsiflexion in swing; they are not interchangeable and each is used only against its own endpoint (§3.3). Supplement S10 gives both corpora in full, including the five further lesion conditions used only as the duration control of §3.3 and the grid's registered seed criterion, and S6 the sub-phase boundaries and the construction of the artefact.

2.4 Analysis and decision rules

The unit of analysis is the lesion condition, not the run: runs within a condition share an optimisation chain and are not independent, giving an effective n of 11, not 59. We fixed four decision rules before reading the results: the noise floor a difference is divided by, the propagation of the survival confound and the fraction of the effect above which it is declared confounding, the per-cycle stationarity a window mean must report, and the centring of every displacement on the unlesioned control chain rather than between groups. Supplement S10 states each in full. We report every difference in degrees, as Cohen's *d*, and in units of the within-condition seed SD, naming the denominator where we use it.

Multiplicity. The reported endpoint is the maximum of a scan. We therefore scored eighty-one candidates: knee, ankle and hip at every five per cent of the gait cycle, plus the peak and trough within stance and within swing, the angle at toe-off, and the range of motion for each joint (§3.1), alongside the sub-phase, video and hip analyses carried from the earlier framing. We selected the severity window after observing where the groups separate, and the endpoint after the scan. No correction is applied and none of these analyses is confirmatory. Two statements in this paper do not depend on the selection: §3.1's count of seven of 81, and §3.4's displacement asymmetry, which is a property of all 81 candidates rather than of the winner.

2.5 Preregistration

We registered the analysis of the mixed grid before reading its results, and that registration does not certify what we report here. All four of its predictions were made about knee angle at initial contact on the mixed grid, so none is a registered test of the endpoint this paper reports. Two are uninformative as registered, one cannot be evaluated because its weakness-dominant cell sits on a grid row that reaches the registration's own four-of-six seed criterion in no cell, and one, P4, is tested in §3.7; two grid elements entered afterwards, the ×0.70 tibialis anterior row and the highest gain column, and Supplement S10 marks which comparisons rest on them. Of 72 registrations, 55 carry a hash sidecar and all 55 still hash to it, and the remaining 17 were never hashed, which is a coverage gap and not a tampering signal. One registration, `PREREG_2ndbody_r229.md`, was edited after its results were produced in order to restore four outcome levels, so the version the run executed is unrecoverable. We report the registration as a record of what was planned and not as assurance about any result, and a reader should treat every analysis here as exploratory. Supplement S7 gives the four predictions in full, the audit of all 97 deposited sidecars, and the post hoc elements.

3. Results

3.1 The candidate scan

Clinical work on this apportionment has most often reported the angle at foot contact, although it is one point on one of three curves and had never been compared with the rest of them. We therefore scored 81 candidate variables on the same runs: knee, ankle and hip at every five per cent of the gait cycle, together with the landmarks a gait report already contains, the peak and trough within stance and within swing, the angle at toe-off, and the range of motion.

A candidate had to clear two criteria before we considered its separation at all. It has to displace the lesioned groups from the unlesioned control by more than twice that candidate's own seed SD, and the two groups' runs have to be disjoint. We also scored a third property, that the two groups be displaced in opposite directions from control, because a sign reversal is one way, though not the only way, for a single number to say which lesion dominates. Seven of the 81 meet all three, and they are not scattered: a contiguous band of the ankle trace in mid-to-late stance (40, 45 and 50 per cent of the cycle), a single point in mid-swing (85 per cent), and the knee near contact (0, 5 and 100 per cent). The factor of two halves the count; at one seed SD fourteen candidates qualify, and on sign alone thirty-three. Knee angle at contact, the endpoint this study began with, is the weakest of the seven.

variable	**control (°)**	**weakness (°)**	**hyperreflexia (°)**	**group difference (°)**	**run gap (°)**	**gap in seed SD**
ankle at 85% of cycle	6.941	8.046	−0.109	8.155	5.033	32.12
ankle at 40%	6.795	7.241	3.954	3.287	1.690	10.66
ankle at 45%	5.863	6.562	2.541	4.021	2.070	9.59
ankle at 50%	2.759	3.529	−0.577	4.106	2.077	6.52
knee at 100%	−7.278	−5.861	−11.316	5.456	2.304	4.36
knee at 5%	−18.521	−17.232	−23.706	6.474	1.827	4.19
knee at contact (0%)	−7.850	−6.492	−11.772	5.280	1.492	3.08

The largest separation in the scan is not in that table. Peak ankle dorsiflexion during swing separates the groups by 8.731° with a run-level gap of 6.265°, which is 76.5 times its own seed SD and twenty-five times the multiple the contact instant achieves. It is excluded because both groups are displaced from control in the same direction, hyperreflexia by −9.161° and weakness by −0.430°, so it carries no sign reversal. We report it as the endpoint nonetheless. A sign reversal is one way of saying which lesion dominates and a large dynamic range is another, and on this dataset the second works better: against the candidates displaced to opposite sides of control, peak swing dorsiflexion separates the groups more completely, survives measurement error better, displaces the unlesioned side less, and is stationary within the window where that band is not (§§3.3, 3.5). It gives up something we report in §3.4, and the loss is not small: it cannot distinguish an isolated weakness from an unimpaired limb. We scanned 81 candidates and report the best (Fig. 3), so every figure for peak swing dorsiflexion below is the maximum of that scan. The count of seven of 81 does not depend on the selection, and the pattern running through all 81 is the subject of §3.4.

3.2 Peak ankle dorsiflexion in swing

	control	**weakness**	**hyperreflexia**	**weak − ctrl**	**hyper − ctrl**	**group difference**
peak swing dorsiflexion (°)	9.555	9.125	0.394	−0.430	−9.161	8.731

Every hyperreflexia run is less dorsiflexed than every weakness run (Figs. 1 and 2). The gap between the nearest cells of the two groups is 6.265°. Against the noise this pipeline generates on the same endpoint, a within-condition seed SD of 0.082° and a protocol artefact of 0.036° (§3.3), that is a margin of 76.5 and 175 times respectively. A hyperreflexia limb reaches 0.39° in swing. It barely clears neutral before the foot comes down, which is the picture Deltombe et al. describe for the equinus that brings these patients to a block in the first place [1]. A weakness limb reaches 9.13° and is hard to tell from the unlesioned model.

At the condition level the difference is +9.117° (95 per cent CI [7.008, 11.226], Welch, df 4.12), computed from five hyperreflexia chain means of 2.310, 1.224, −0.645, −1.023 and −1.826° against six weakness means of 8.734, 9.045, 9.145, 9.168, 9.263 and 9.396°. That interval is not a sampling interval. We designed the five hyperreflexia conditions as a dose series, so adding chains at the same doses would not narrow it. Permuting the eleven condition labels exhaustively, all C(11,5) = 462 ways, places the observed split first at p = 0.0022, which is 1/462 and the smallest value the design can return. Leave-one-condition-out classification is correct on 59 of 59 runs against a 61.0 per cent majority-class baseline. We computed both figures for a variable we had chosen as the maximum of an 81-candidate scan. Therefore, we repeated each with the selection inside the procedure: a max-T permutation that rescans all 81 candidates within each of the 462 relabellings returns the same p of 0.0022, and a nested classification that chooses the variable from the ten training conditions inside each fold picks the same variable in all eleven and returns the same 59 of 59. The permutation cannot fall below 1/462 on an eleven-condition design and returns exactly that for each of the thirty candidates whose groups are disjoint, so it does not discriminate among them. The classification is not saturated in the same way: leave-one-condition-out fits its threshold on ten conditions. A held-out condition can fall the wrong side of it even where the groups are globally disjoint, and knee angle at contact, which is disjoint, returns 0.983. What separates the reported endpoint from the rest is the size of its gap against its own noise. Supplement S9 gives the null distributions and the per-fold selections.

We take the endpoint in swing because that is the one place the phase of the model's ankle matches a normal one. Its value there does not: the unlesioned peak is 9.56° where a normal swing peak is 0 to 5°, which is discussed in §4.4 and bears on every displacement measured from this control. What holds in swing and nowhere else is that the model's ankle moves as a normal ankle does. The unlesioned trace leaves stance at 59.5 per cent of the cycle, reaches its plantarflexion minimum of −19.8° at 63 per cent, and peaks in swing at 81 per cent. Of the four landmarks a clinician would recognise, only that last one falls where a normal adult ankle reaches it: the loading-response trough is early and shallow, and the terminal-stance peak, which a normal ankle reaches near 45 to 50 per cent, comes early here as well. §4.4 returns to that, and it is why we report no stance-phase variable.

3.3 Controls

The separation could arise without the lesion producing it: from drift under continued optimisation, from the difference in survival between the groups, from a bilateral re-solution instead of a lesion-side displacement, or from the choice of admissibility criterion. Re-running a chain through further optimisation rounds at unchanged severity displaces this endpoint by at most 0.036°, against 0.586° for knee angle at contact (Supplement S5), and the endpoint's displacements do not grow with chain depth, although the knee's do, at roughly 0.15° per round, which is why its floor and the endpoint's are not interchangeable. We report every effect below as a multiple of 0.0358°, printed here as 0.036°. Hyperreflexia displaces the endpoint by −9.161°, 256 times that floor, and weakness by −0.430°, twelve times it. Both lesions move the ankle the same way and differ only in how far, so the endpoint reports how overactive the calf is and nothing about which side of normal the limb has ended up on.

Simulation duration is the confound we cannot close. Every weakness run reaches the 20 s horizon and every hyperreflexia run ends between 9.81 and 17.85 s. The regression our decision rules prescribe gives a slope of −0.041 $°\cdot s^{-1}$ and propagates 0.298° over the 7.28 s difference, 3.4 per cent of the effect, but that is not a bound: the rule requires within-condition variation in end time, which no weakness condition has, so all five contributing conditions are hyperreflexia conditions and the slope is extrapolated into a regime containing none of them. Across those five the standard error of the mean is 0.102, and we read the regression as 0.30 ± 0.74° at one standard error, or 3 ± 24 per cent of the effect at 95 per cent. Pooling all runs across both groups instead gives 1.101 $°\cdot s^{-1}$ and propagates 92 per cent of the effect, but that estimator is circular, since pooling across groups recovers the group difference the regression is meant to test.

The archive supplies a more direct control. Separate chains carry a plantarflexor-weakness lesion, which is neither of the two groups and involves no stretch reflex, and many of their runs die early. Seventeen end between 9.93 and 17.56 s, inside the hyperreflexia group's own band, and their swing ankle is normal: mean 9.572°, displaced 0.017° from the unlesioned control, which is 0.47 times this endpoint's artefact floor, with a range of [9.512, 9.673] disjoint from the hyperreflexia group's [−1.974, 2.404] by 7.108°. They fail to produce the knee signature too, at −7.488° against control's −7.850° and hyperreflexia's −11.772°. Short survival by itself does not make a limb look hyperreflexic.

That control has a weakness of its own, which a second one addresses. Its lesion is plantarflexor weakness, a manipulation of the same muscle group whose overactivity is claimed to drive the effect, so it does not establish that a limb falling for an unrelated reason reads as unimpaired. One archived condition supplies that. Scaling hamstring force to 0.60 on the same model and controller produces a lesion in muscles that cross the hip and knee and not the ankle, and every seed falls. Five are readable under the corpus conventions, and they fall earlier than any hyperreflexia run, between 6.46 and 9.03 s, so they end before criterion G opens and are measured over the four cycles they complete rather than five: this is not a duration-matched comparison but an asymmetric one, in the direction that favours the objection. Their swing ankle reads 9.470°, displaced 0.086° from the unlesioned control, which is 2.4 times the protocol artefact and 107 times smaller than the hyperreflexia group's 9.161°, and their range is disjoint from the hyperreflexia group's by 7.025°. At the knee the same runs are less clean, reading −9.398° at contact against control's −7.850° and hyperreflexia's −11.772°, so a hamstring lesion does move that endpoint about a third of the way. Whether the hyperreflexia group's own loss of balance adds something on top of the reflex we still cannot say, and no design here could. Supplement S4 gives this control's selection check and the four remaining controls in full.

Of those four, the per-cycle group difference keeps one sign across the five analysed cycles with a spread of 0.276°, 3 per cent of the window mean, against 14 per cent for knee angle at contact and 55 per cent for the two-sided ankle band of §3.1. On the unlesioned right ankle the group difference is 0.047°, 0.5 per cent of the lesioned side, where knee angle at contact carries 35.5 per cent of its group separation on the unlesioned side. Walking speed differs between the groups by 0.056 $m \cdot s^{-1}$ and propagates 0.032°. Raising the admissibility criterion from 9.73 s to 13.00 s removes 12 of the 23 hyperreflexia runs and none of the 36 weakness runs while the run-level gap rises from 6.265 to 6.302°.

3.4 Displacement and grading across the scan

Against the unlesioned control, plantarflexor hyperreflexia displaces peak swing dorsiflexion by −9.161° and dorsiflexor weakness displaces the same variable by −0.430°, a ratio of 21 to 1. That ratio is particular to this variable, although the asymmetry itself is not: across the 81 candidates the median displacement is 0.49° for weakness against 1.90° for hyperreflexia, weakness reaches 3.95° at the knee in early swing, and on ten of the 81 it displaces more than hyperreflexia does. The direction and the order of magnitude generalise; the factor of 21 does not.

Three variables order the six tibialis anterior scalings at $|\rho| \geq 0.8$ over spans wider than one clinical measurement error: the ankle at 75, 80 and 90 per cent of the cycle, at $\rho = +0.822$, +0.900 and −0.820 across 2.870, 2.208 and 2.272°. Applying the same test to the hyperreflexia group, we find 25. Neither count is fixed. Which published measurement error serves as the threshold, and whether the rank correlation uses mid-ranks on a tied dose vector, move them: across three defensible thresholds and three rank conventions the tone count ranges from 14 to 28 and the weakness count from 1 to 7, and their ratio from 3.7 to 14.0 without ever inverting. That upper bound is a property of the nine readings and not of every threshold: sweeping a single common threshold instead, the weakness count reaches zero above about 3° while the tone count is still 14, so the ratio there is unbounded rather than bounded by 14 (Supplement S8). We therefore report the asymmetry as a direction and an order of magnitude, roughly fourfold at its weakest and fourteenfold at its strongest, and not as a pair of counts. The denominator is not 81 independent opportunities either, since adjacent samples of one curve are strongly correlated, so the counts index how much of the cycle each lesion is legible over: calf overactivity over between a sixth and a third of the sagittal variables a gait laboratory produces, and weakness over a small fraction of that.

The three variables that grade weakness grade tone more strongly at the same phase, their hyperreflexia spans of 6.308, 6.055 and 4.148° standing against weakness spans of 2.870, 2.208 and 2.272°, larger by 2.2, 2.7 and 1.8 times. A reading taken there moves for two reasons at once.

Weakness depth is not recovered from a reading that does not already know the tone. On the mixed grid it is recovered neither by knee angle at contact, nor by peak swing dorsiflexion, nor by the two together, nor by the two with their interaction: every leave-one-weakness-level-out coefficient of determination is negative, so each does worse than reporting the mean. The fold is demanding but passable, since the parallel leave-one-gain-level-out folds return R^2 between 0.64 and 0.82 for reflex gain. Reflex gain is recovered from one reading of one variable at $\rho = -1.000$ across the five gain levels and -0.964 over the 23 hyperreflexia runs, with a leave-one-condition-out calibration error of 0.0062 in delivered gain, 4.5 per cent of the series' range; that condition-level correlation is 1.00 by construction on five ordered levels, and what the run-level figure adds is that the ordering survives seed-to-seed dispersion. We ran that regression on four feature sets and not on all 81, and tested one two-stage estimator, which recovers gain at $R^2 = 0.946$ and then weakness from the knee residual at $R^2 = 0.129$.

The failure has a mechanism. Holding reflex gain fixed on the mixed grid and varying tibialis anterior scaling, the weakness levels are ordered perfectly at delivered KV 0.025 ($\rho = +1.000$, span 1.201°), flat at KV 0.050 ($\rho = -0.400$, span 0.191°), and ordered perfectly with the opposite sign at KV 0.100 ($\rho = -1.000$, span 1.449°) and KV 0.220 ($\rho = -1.000$, span 0.753°).

Whether weakening the dorsiflexors raises or lowers peak swing dorsiflexion therefore depends on how hard the plantarflexors pull against them, with the crossover near KV 0.050. An estimator blind to the tone cannot use a signal whose sign it cannot predict. Supplement S14 gives the dose response in full.

A gait recording can report how overactive the calf is. It reports how weak the dorsiflexors are in three of 81 places, over spans a half to a third the size of the tone signal at the same phase, and, once tone is present, only in a direction that depends on how much of it there is. In any limb with calf overactivity a strength measurement is therefore the only available source of the second quantity.

3.5 Detection, discrimination and grading under measurement error

Three separable claims can be made for this variable: that it detects a lesion, that it distinguishes the two, and that it grades either. We scored each on the noise-free simulation and again under the measurement model of §3.6, at 100 Hz with yaw ≤ 10° and event error ≤ 1 frame, over 60 noise realisations of the full primary dataset (Fig. 5).

at 3° of per-frame jitter	**detect hyperreflexia**	**detect isolated weakness**	**distinguish the two**	**grade tone (ρ)**	**grade weakness (ρ)**
peak swing dorsiflexion	1.000	0.512	0.995	−0.902	+0.227
ankle 40 to 50% window	0.988	0.650	0.994	−0.849	−0.330
ankle at 85%	0.963	0.537	0.983	−0.792	−0.138
knee at contact	0.766	0.535	0.939	−0.672	−0.218

Distinguishing the two lesions survives measurement error on every candidate, best on variables that take a peak or average a window, though for opposite reasons: a window mean divides zero-mean jitter by the root of its sample count, while a maximum amplifies it (Supplement S3) and survives on the size of its clean gap. Grading tone survives on the ankle and degrades sharply on the knee. Grading weakness survives nowhere: the three variables that grade it on noise-free traces were put through the same model separately and none clears the 0.8 criterion at 3° of jitter, so under this study's own measurement model that count falls from three to zero, and §3.4's counts should be read as noise-free throughout. The endpoint is not best in the two weakness columns, and in both the reason is §3.4's asymmetry rather than a property of the variable.

Two conventions govern the table and Supplement S1 sets out both with the noise model's own limits. The correlations are signed, so a sign reversal counts against a variable and the small values in the weakness column are no recovery rather than partial recovery. The two detection columns are sensitivities and not accuracies, since no control run is classified in producing them: the endpoint's 1.000 for detecting hyperreflexia becomes 0.687 once the six control runs are classified as well.

A tone score follows from the same readings, and it is ordinal and not absolute. The single measurement error this study uses throughout is 1.77°, the between-session standard error for peak ankle angle in swing published for a post-stroke cohort by Kesar et al. and inverted here through that paper's own definition of minimal detectable change [15]; its authors state that their figures may not be used to interpret overground gait, and Supplement S1 gives the derivation and the within-session figure of 0.32°. Taking T as the control mean minus the patient's reading, a threshold at $T = 4.02°$ separates the two groups with 3.13° to spare, 1.8 times that error. Under noise the five T values move down by up to 1.505°, which exceeds three of the four gaps between adjacent levels, because taking a maximum biases the reading upward and the flatter hyperreflexia peak is biased more; the group difference compresses from 8.731 to 7.431°, 85 per cent of its clean value. The ordering itself is unaffected, which is why the rank correlation of $\rho = -0.902$ survives, but T ranks severity without placing a patient on the series, and a recording with 3° of jitter would place one up to three levels low. Supplements S1 and S3 give the five T values, the two published measurement errors and the rectification measurement.

The grading is bounded twice over, and the two bounds are independent. Only one of the four adjacent gaps exceeds the 1.77° between-session error, while the series' full span of 4.14° is 2.3 errors wide; and the rectification drift is of the same size as the gaps it would have to resolve. Errors of this size are "generally not small enough to be ignored during clinical data interpretation" [24]. The variable supports a coarse contrast, substantial calf overactivity against little, and not a continuous or absolute tone scale. These figures carry no between-patient variance and are not clinical accuracies; what the effect is worth against the spread a patient population presents is set out in §4.1.

3.6 The measurement model and the recording modality

The noise model behind §3.5 is optimistic in four respects that Supplement S1 sets out: the jitter is independent and Gaussian, the yaw term is inert on a planar trace, event detection is exempted by taking the swing window from the noise-free ground reaction force where a real pipeline must estimate it [23], and it models one recording rather than two, so the error governing a patient's reading is the 1.77° of §3.5 and not anything in this grid.

Published sagittal errors for markerless video are, for the knee, 5.6° MAE in unimpaired adults [21] and 4.0° post-stroke [22]. For the ankle, this paper's endpoint, they are worse: 7.4° MAE with a cross-correlation of 0.74 to 0.78 against 0.97 to 0.98 for the hip and knee [21], and 6.3° post-stroke at the ankle from sagittal video [22], comparable to the whole 8.731° effect and larger than the 4.14° tone series. Video-based pose estimation cannot support the grading proposed here, and marker-based capture is the only modality these results transfer to.

3.7 The registered prediction, tested at both endpoints (Fig. 4)

The registered prediction that bears on apportionment was made at the contact instant, and it held there. It predicted that different lesion combinations would produce overlapping readings, and at the knee they do: among the six pairs the registration covers, the closest two differ by

0.302°, less than that endpoint's artefact floor, so two limbs with different amounts of each lesion read the same. At peak swing dorsiflexion the answer depends on which pairs are counted. Among the 96 pairs that differ in reflex gain the closest differ by 0.833°, 9.3 times that endpoint's grid floor of 0.089°, but of the 24 that hold gain fixed and vary dorsiflexor strength alone, four fall below the floor and the closest at 0.044°. The endpoint therefore does not separate two limbs whose calves are equally overactive and whose dorsiflexors differ, which is §3.4's asymmetry restated one pair at a time; the earlier claim that no pair falls within one seed SD is withdrawn.

The two endpoints differ in resolution and not in excursion. Over the grid's gain range of 0.025 to 0.220 the gain axis moves the swing peak by 8.210° and the contact knee by 8.113°, on noise floors differing about sixfold, and neither endpoint rescues the weakness axis (§3.4). Supplement S2 gives the dose response, the sub-phase sign reversal at the knee, and the difference in the two populations of pairs.

3.8 Localisation of the separation

Because the controller is re-optimised for every lesion, the joint at which a difference appears does not identify the muscle responsible. Scored against the sham artefact, the unlesioned-side control for knee angle at contact fails from delivered KV 0.100 upward, and at the two deepest gains the unlesioned knee is displaced from control by 3.44 and 4.01°, carrying 35.5 per cent of that endpoint's group separation (§3.3), so a re-optimised body redistributes a local deficit and a single-joint reading registers the redistribution along with it. That a distal constraint displaces proximal kinematics is not itself surprising, since imposing equinus mechanically on unimpaired adults alters knee kinematics during gait [19].

The reported endpoint is the least exposed to this. Its displacement of the unlesioned side is 0.5 per cent against the knee's 35.5, both scored at the unlesioned limb's own contact (§3.3), and scored instead at the lesioned limb's contact the knee figure is 30.8 per cent while the ankle figure is unchanged. What it registers is therefore almost entirely a lesion-side displacement, although it grades calf overactivity as a marker and not as a direct observation of the calf. We searched for a mechanism behind the redistribution and found none whose effect exceeds the protocol artefact; one candidate, a left-minus-right hip asymmetry, is examined in Supplement S11 and is not reported here.

Two results do not survive as stated. The ankle-at-contact null fails the noise-floor and centring rules of §2.4, so it is not certified by this design, and knee angle at contact loses lesion-side specificity at the deeper gains where the unlesioned knee moves as well (§3.3). No control here can reach a third question, whether the separation follows from lesion type or from the lesion's effect on stability.

4. Discussion

4.1 Interpretation

Plantarflexor overactivity is legible over between a sixth and a third of the sagittal variables a gait laboratory produces, and dorsiflexor weakness over a fraction of that, moving the reading 2.2, 2.7 and 1.8 times less far than tone does at the same phase where it is legible at all (§3.4). A gait recording grades one component of a spastic equinovarus foot well and the other faintly, and that is unlikely to be an instrumental limitation a better laboratory would remove.

The instant appears to matter more than the joint. At foot contact the ankle does not separate the groups and the knee does, which is the comparison we first made and it stands, but it was never the whole of clinical practice: peak dorsiflexion in swing is already a reported variable, and in a 34-patient chronic stroke cohort it separated the contact-type groups more strongly than the angle at contact did ($p < 0.001$ against $p = 0.003$) [5]. Extended to the whole cycle the ordering reverses. The best variable in the study is at the ankle in swing, exceeding knee angle at contact by a factor of twenty-five in gap-to-noise and the best knee variable anywhere in the scan, the knee at 25 per cent of the cycle at 17.4 seed SD, by a factor of four.

Gap-to-noise is a property of the pipeline and only part of it is likely to transfer. Against the between-patient spread the ankle gains twice over: the effect is 1.7 times larger and the spread at its landmark 1.5 times smaller, 5.55° against 8.57°, both reconstructed here from the two subgroups of ref [5] and not reported there. With a single measurement's error this gives a total SD of 5.82°, against which the 8.731° effect is $d = 1.50$ and misassigns about 22.7 per cent of single readings, where knee angle at contact gives $d = 0.59$ and misassigns 38.4 per cent on its own between-patient SDs and a single-measurement error of 2.62° that we re-derived from [13], which measured 50 healthy subjects rather than a stroke cohort. The one published reference set of 138 able-bodied adults [25] reports no swing-phase ankle SD, so it cannot corroborate the 5.55°. A two-group comparison at the ankle would therefore need roughly a sixth the sample it needs at the knee, which is a property of the discriminability and not a sample size for the correlational study §4.3 proposes. These data support measuring the ankle in swing instead of at contact; they do not support moving from the ankle to the knee. Supplement S15 gives the derivation, the cohort's exclusions, and the caveat attaching to the knee comparator's reliability figure.

A simulation is the instrument this question needs, because the apportionment requires both lesion magnitudes in one limb at once and the only procedure that supplies either is the diagnostic block, which is invasive, removes one component and cannot calibrate the other. Supplement S13 sets out that argument.

4.2 Scope of the reading

The lesion imposed here is a velocity-dependent stretch reflex, which is spasticity in the strict sense, and it is not the only plantarflexor abnormality a hemiparetic limb carries. In eleven hemiparetic adults, five of whom had undergone tibial neurotomy abolishing plantarflexor spasticity, plantarflexor cocontraction during swing was present in the operated group and larger than in the unoperated one, soleus coefficient 1.48 ± 0.2 against 0.95 ± 0.3, and began before any stretch [26]. Against diagnostic nerve block the same conclusion is reached independently: improvement in swing dorsiflexion after block was associated with soleus cocontraction rather than with Tardieu-graded spasticity, and although that association is not itself significant (OR 4.72, 95 per cent CI 0.86 to 26.04), the absence of any association between Tardieu-graded spasticity and soleus overactivity during swing is unambiguous in the same study [3]. Ref [26] reports no ankle kinematics separately for its two subgroups, so the proposition that abolishing spasticity leaves the dorsiflexion deficit unchanged is asserted there rather than tested, and its operated subgroup's median comfortable velocity was the higher of the two, 0.81 against 0.50 $m \cdot s^{-1}$.

That work bounds the interpretation of this variable without contradicting the result. A velocity-dependent stretch reflex acting alone displaces the variable by 9.161° and grades monotonically with its own strength; what cannot be established from a model containing no other plantarflexor abnormality is that a low reading in a patient is produced by that reflex rather than by cocontraction or by shortening. The endpoint should therefore be described as grading calf overactivity, not spasticity. For the decision it is offered to serve, part of that is not a retreat, since overactivity from a stretch reflex and overactivity from cocontraction are both treated by chemodenervation or neurotomy. The retreat is larger, because this model contains no muscle–tendon shortening: a limb whose calf is short rather than overactive would read low here as well, and shortening is treated by stretching or lengthening, a distinction the assessment pathway is organised around [1] and one shown to change the treatment chosen in individual patients [20]. The reading separates plantarflexor abnormality of any origin from dorsiflexor weakness, and a clinical protocol using it would still need the passive assessment that separates overactivity from shortening.

4.3 A patient study

This study offers a negative with a consequence and a calibration. The negative is that no sagittal joint angle, alone or in combination, reports dorsiflexor weakness over the range examined once measurement error is applied (§§3.4, 3.5). An instrumented assessment that measures gait and infers strength from it is measuring one component twice, so the two-predictor design a patient study should use, plantarflexor tone and dorsiflexor strength entered together, is required by the second predictor being unobtainable from the first's data rather than by statistical prudence.

The calibration is of a variable clinics already record. Peak ankle dorsiflexion in swing orders reflex gain at $\rho = -1.000$ across the five gain levels; against the between-patient spread of a chronic stroke cohort it misassigns about one reading in four, its five-level series resolves only one adjacent pair above the between-session measurement error, and the rectification drift of §3.5 is the size of the gaps it would have to resolve. It supports a coarse contrast, a limb with substantial calf overactivity against one without, and not a graded tone score for an individual. Marker-based capture is required, since reported ankle errors for video-based pose estimation reach 7.4° in unimpaired adults [21] and 6.3° post-stroke [22] (§3.6).

The study we would run next is cross-sectional with three measurements per limb: instrumented gait giving peak dorsiflexion in swing; a graded measure of calf overactivity, ideally against a diagnostic block in the subgroup receiving one; and dorsiflexor strength by dynamometry. The closest published design is a between-day reproducibility study of paretic-limb sagittal kinematics in 26 chronic stroke patients over three weekly sessions [14], whose full text we could not obtain and from which no value is quoted here; a study of that shape with the two predictors above added is what we propose. The relation to establish is whether the gait variable tracks graded overactivity once strength is held constant, which is what this simulation predicts and which the co-occurrence of the two components in patients ([4];

Supplement S13) makes impossible to read off an unadjusted correlation. Detecting a partial correlation of 0.4 with strength partialled out needs about 50 limbs, and 0.3 needs about 90; one session's measurement error attenuates such a correlation by 4 to 6 per cent, which is not the limiting consideration at those sample sizes.

4.4 Limitations

The model and its lesions. No convention governs how spasticity and weakness enter a musculoskeletal simulation: gains chosen arbitrarily [9], force and tendon slack scaled per condition [10], hyperreflexia set against plantarflexor weakness in two dimensions [11], and in personalised cerebral-palsy models impairments of known magnitude explaining on average 17 per cent of the kinematic deviation and never more than 39 [12]. Our study sits inside that spread, and the asymmetry we report rests on how much reserve this model's tibialis anterior retains in an unloaded swing limb, which no internal control reaches. Ours is also a shallow weakness range: Ong et al. model mild, moderate and severe plantarflexor weakness at 25, 12.5 and 6.25 per cent of maximum isometric force [10], where our deepest is 60 per cent and the primary series runs from ×0.80 to ×0.95. We could not go deeper and keep a walking model under this controller. The unlesioned ankle is also only approximately physiological, dorsiflexing to +10.7° by 15 per cent of cycle where a normal one is near neutral, and reaching a swing peak of 9.56° above the normal range of 0 to 5°. Taking the endpoint in swing avoids the phase problem and not the value problem, and the value problem has a consequence we can state but not remove. Because the control sits above the normal range and the hyperreflexia group sits at 0.394°, this model's lesioned limb falls inside the normal range at this landmark while its unlesioned limb falls outside it. Every displacement we report is a difference between two readings of this model and is unaffected. The tone score is not: T is defined as a control mean minus a patient's reading, and a clinic would use its own unimpaired reference rather than this model's 9.555°, which would shift all five T values by five to nine degrees. The threshold of 4.02° and the 3.13° of margin are therefore internal to this model, and a clinical version of the score would have to be re-established against a real reference before any of the numbers in §3.5 could be used.

What the design cannot separate. Predictive simulations stop walking under a deep enough deficit, which is a property of this class of model rather than of this study [10], and here that coupling falls along the lesion axis: none of the 23 admitted hyperreflexia runs reaches the horizon and all 36 weakness runs do, so no group comparison at matched survival exists. The within-condition duration regression gives 3 ± 24 per cent of the effect at 95 per cent and is an extrapolation rather than a bound; the seventeen non-reflex fallers show that short survival is not sufficient for the effect; neither settles whether the hyperreflexia group's own loss of balance contributes to it. Separately, we imposed the two lesions independently, and patients do not present that way ([4]; Supplement S13). In a correlated population a variable insensitive to weakness will still track it through its association with tone, so a low swing dorsiflexion cannot be attributed to one component from the reading alone. Since the treatment decision does turn on which component is present, this limits the reading's clinical use and not merely the external validation of the specificity claim.

The status of the result. We chose the endpoint after seeing the data, from a scan of 81 candidates; we apply no correction for multiplicity, treat no analysis as confirmatory, and report the endpoint's figures as the maximum of that scan. The statement that survives the selection is §3.4's, which holds across all 81. The method is not new: a reflex-controlled model with graded deficits and per-deficit re-optimisation is established [10], as is grading a spindle-feedback gain inside a gait model [9], which also reports the direction of the swing effect. What neither did is cross two lesions of different kinds and score a variable on what it says about each. And every figure bearing on a clinic combines this study's effect sizes with published between-patient SDs and reliabilities. We measured nothing here in a patient.

The insensitivity to weakness holds in four respects only: for tibialis anterior scaling between ×0.60 and ×1.00, for sagittal joint angles, for a design in which walking speed is nearly matched, and for this model and controller. Outside them it is untested, and inside them it is contested: Chisholm et al. report, in 55 independently ambulating sub-acute stroke survivors, that peak dorsiflexion during swing was related to isometric dorsiflexor force at $r = -0.32$, $p = 0.039$ [28], which is an association in patients at this study's own endpoint of a size this model calls faint. Only the published abstract of [28] could be retrieved, so we cannot check whether that cohort's dorsiflexor forces overlap the range tested here; Supplement S12 sets out what follows on either reading. Supplement S12 gives each limitation with its workings, including the patient results that bound the asymmetry and the three pathways by which weakness may still reach the endpoint.

5. Conclusions

Peak ankle dorsiflexion during swing separates plantarflexor hyperreflexia from dorsiflexor weakness completely at the run level and grades the reflex at $\rho = -1.000$ across the five gain levels, and it does both under a simulated motion-capture error that leaves knee angle at contact substantially degraded. Gait laboratories already record the variable, so what this study offers is its calibration rather than a new measurement. Both figures are the maximum of an 81-candidate scan and so an upper estimate of what a pre-specified choice would have found.

The same dataset sets the bound. Of the 81 candidates, between 14 and 28 grade the reflex usably against 1 to 7 for dorsiflexor weakness, a ratio that never inverts and that stays between 3.7 and 14 across those nine readings, although it is unbounded above under a stricter common threshold. Under this study's own noise model the weakness count falls to zero for the three variables tested. Where both lesions are present, no variable and no combination we tested recovers how weak the dorsiflexors are without first knowing the tone, because the weakness signal reverses direction across the gain range. Half the apportionment a clinician needs is answerable from gait; the other half requires a strength measurement gait cannot supply.

None of the simulation-side figures is a clinical accuracy. Against the spread a chronic stroke cohort presents at this landmark the same effect misassigns about one reading in four, a figure that already contains a single measurement's error and rests on the model's 8.731° transferring unchanged to patients. Better instrumentation would not reduce it, because the limit is set by the spread between patients. Whether it improves on what a clinician achieves unaided is a comparison nobody has made, and the one study setting clinical assessment against diagnostic nerve block found the clinical data unassociated with the instrumented findings [3].

Both lesion magnitudes are known here and in no patient. The reading grades calf overactivity and not the mechanism producing it [3, 26]. The hyperreflexia group falls before the horizon while the weakness group reaches only ×0.80, against a population whose median dorsiflexor manual muscle test is 1 of 5 [4], so the two series are matched neither in survival nor in severity. And the insensitivity to weakness holds where walking speed is nearly matched. Dorsiflexor strength is the strongest single determinant of gait velocity in chronic hemiparesis [27], and this endpoint depends on speed at -0.563 °/($m \cdot s^{-1}$) (Supplement S4), so across the 0.2 to 1.2 $m \cdot s^{-1}$ range a stroke cohort walks at the indirect path from weakness through speed carries about 0.23°, comparable to the entire direct effect of 0.430°.

Within those conditions this delivers a calibration of an existing measurement, a bound on what any sagittal angle can carry, and a patient study whose second predictor is now known to be necessary rather than merely prudent. These results do not support using the variable as an absolute measure of tone in an individual patient.

Figures

Fig. 1 Ankle angle over the gait cycle for the three groups, six optimisation seeds per condition except at the two deepest reflex gains, which carry four (§2.1). **A** the whole cycle; **B** swing, where the reported endpoint is taken. Bands are ±1 SD across seeds *of the pooled group*, not across subjects. Dorsiflexion is positive and the dotted line marks mean toe-off at 59.5 per cent of the cycle. The unlesioned trace reaches its swing peak at 81 per cent, the one landmark of the four that falls where a normal adult ankle reaches it, and plantarflexes to −19.8° near toe-off, and shows a loading-response trough that is early and shallow. It also dorsiflexes to +10.7° by 15 per cent of the cycle, where a normal ankle is near neutral, and its terminal-stance peak is correspondingly early: the non-physiological features recorded in §4.4 and the reason no stance-phase variable is reported.

Fig. 2 Peak ankle dorsiflexion in swing, cell by cell, with the control chain at the top and the two lesion series below. Each point is one optimisation seed. The shaded band is the run-level gap: every hyperreflexia cell lies below every weakness run, by 6.265° or 76.5 within-condition seed SD. Note that the weakness series sits 0.43° from control while the hyperreflexia series sits 9.16° from it, on the same side. This is the asymmetry of §3.4, visible here as the two lesions standing unequally far from normal instead of on opposite sides of it.

Fig. 3 The scan of §3.1. Cell-level gap, in units of each candidate's own within-condition seed SD, for the knee, ankle and hip sampled at every five per cent of the gait cycle. Rings mark the candidates that are also two-sided, that is, that displace the two groups to opposite sides of the unlesioned control by more than twice that candidate's own seed SD (§3.1; on sign alone the count is thirty-three rather than seven). The star is the reported endpoint, which is a landmark rather than a fixed phase and so is plotted at its mean phase. The vertical scale is clipped below −2 SD. **This figure is the selection**: the reported endpoint is its maximum, and every figure derived from that choice is an upper estimate.

Fig. 4 The mixed grid, in which both lesions are present at independently known magnitudes. **A** peak swing dorsiflexion against delivered reflex gain, one line per tibialis anterior scaling: the lines move together and span 8.21°. **B** the same values against tibialis anterior scaling, one line per gain: the lines are nearly flat, span at most 1.45° within any gain column, and change direction across the gain range. The contrast between the two panels is the paper's principal result.

Fig. 5 What a simulated motion-capture error leaves of two of the three claims. **A** leave-one-condition-out accuracy in distinguishing the two lesions; **B** Spearman correlation between the reading and delivered reflex gain. The horizontal axis is per-frame Gaussian jitter at 100 Hz with yaw ≤ 10° and event error ≤ 1 frame. The third claim, grading weakness, is not plotted because it does not rise above chance at any noise level. **These are simulation resolutions and contain no between-patient variance**; the corresponding single-patient figure is the 22.7 per cent misclassification of §4.1.

Declarations

Ethics approval and consent to participate. Not applicable; no human or animal subjects.

Consent for publication. Not applicable.

Availability of data and materials. All simulation outputs, preregistrations with hashes, analysis scripts, correction records and the defect register are deposited at [REPOSITORY DOI]. Every figure and statistic in this manuscript is reproducible from the archived outputs without a simulator licence, with two exceptions the reader should know: the Hyfydy licence under which these simulations were run expired on 27 August 2026. No new simulation can be added to the dataset; and 16 of 1129 archived run directories lack a replayed output and contribute to no analysis here.

Use of large language models. A large language model wrote and revised the analysis scripts, drafted and revised this manuscript, and ran the adversarial review rounds recorded in the deposited defect register. The research question, the choice of what to claim, the decisions to withdraw particular claims, and the decision to submit were the authors'. The model is not an author and meets no authorship criterion.

An earlier version of this declaration stated that every number had been checked against its container and that every value attributed to a cited work had been verified against the retrieved source. Neither was true when it was written, and both are corrected here. The self-check in place at that time tested for the presence of strings and performed arithmetic between constants written into the checker; it opened no container. An automated binding check was added afterwards and is deposited with the analysis: of the 222 numeric values of three or more decimals in the manuscript and supplement, 199 match a value in a deposited container whose path shares a term with the claim and 23 do not, and that check lists all 23. It is a token match and not a semantic one, and several of the 23 are values that do appear in a container at higher precision than the text prints. Values of fewer than three decimals, which include several of the clinical figures, are outside its scope. On citations, values attributed to cited works were checked against retrieved sources, and that audit removed three figures previously attributed to [28], for which only the published abstract could be retrieved. Cases where a source proved internally inconsistent are stated in the text.

The authors take full responsibility for the content, including the parts the model drafted and including the two claims corrected in the paragraph above.

Competing interests. The authors declare no competing interests.

Funding. [TO BE COMPLETED]

Authors' contributions. [TO BE COMPLETED]

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